Multiple testing

The problem with multiple testing: An example

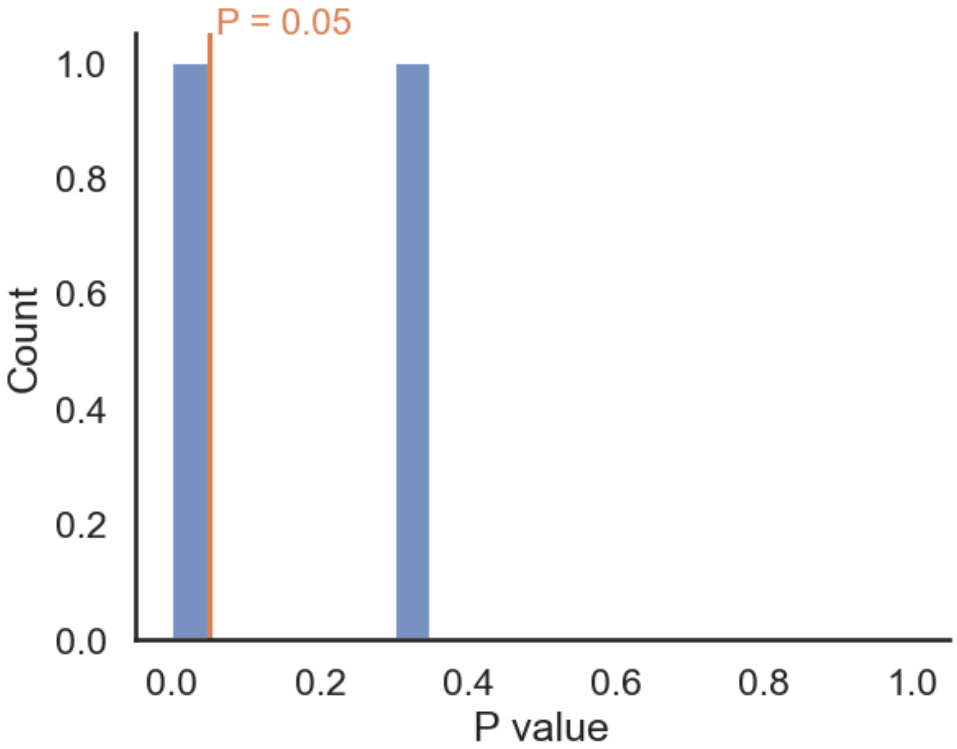
The problem with multiple testing

The problem with multiple testing

Before	After	Difference
144	144	0
143	146	3
144	146	2
145	145	-1
146	147	1
146	147	1
143	145	2
144	148	4
145	148	3
145	143	-2

A second experiment in a different lab

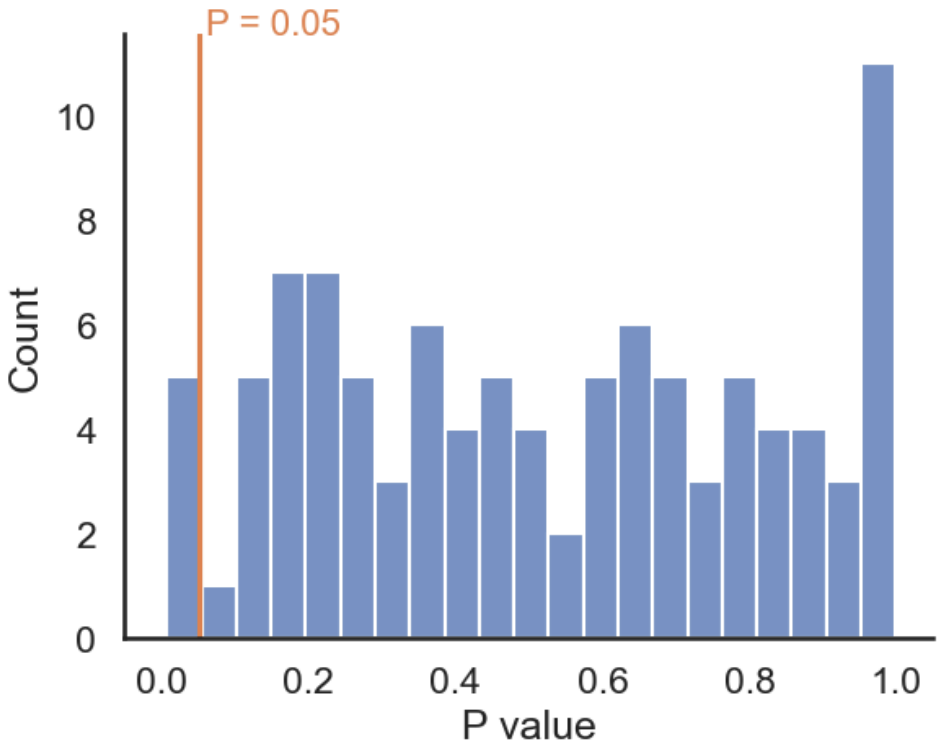
- $P = 0.031$, incompatible, a significant effect



The problem with multiple testing

Consider 100 independent experiments

- In 100 independent experiments in which the null hypothesis is true, five are significant
- These are called false positives
- When the null hypothesis is true, 1 in 20 (5%) significance tests result in a false positive
- If your analysis requires multiple testing, you need to reduce your risk of making false claims



Multiple testing

Occurs in any scientific study where many statistical hypotheses are tested simultaneously on the same dataset

- Multiple treatments:
  - Comparing plant biomass with different fertilisers compared to a control of no fertiliser
  - Bird species richness is compared between old-growth forest, logged forest, pine plantation and urban park (6 pairwise comparisons)
  - Multiple drugs screened simultaneously for liver toxicity
- Multiple outcomes:
  - A drug trial evaluating a new medication might track its impact on many different health endpoints (e.g., blood pressure, cholesterol, heart rate, kidney function, and stroke risk)
  - Differential expression of thousands of genes/proteins between a control and treatment

Two main methods to control for multiple testing

- The Holm method (published by Sture Holm in 1979)
  - Use when false positives are costly
    - Clinical trials, safety decisions, regulatory settings
    - A small number of pre-specified comparisons
- ~~Bonferroni method~~
  - No longer recommended as too conservative (rejects too many true positives). Use Holm instead

Two main methods to control for multiple testing

- The Benjamini–Hochberg method (published in 1995)
  - Use when you're screening and can tolerate some false leads
    - Genomics, proteomics, neuroimaging (hundreds–thousands of tests)
    - Exploratory/discovery phase where follow-up validation will catch false positives

Holm: Comparing treatments to a control

- Plants were grown with
  - no fertiliser (control)
  - fertiliser A (treatment 1)
  - fertiliser B (treatment 2)
- Their dry mass (in grams) was measured after 8 weeks.



Read in and print the data

```
pg = pd.read_csv('../Datasets/plant_growth.csv')
pg
```

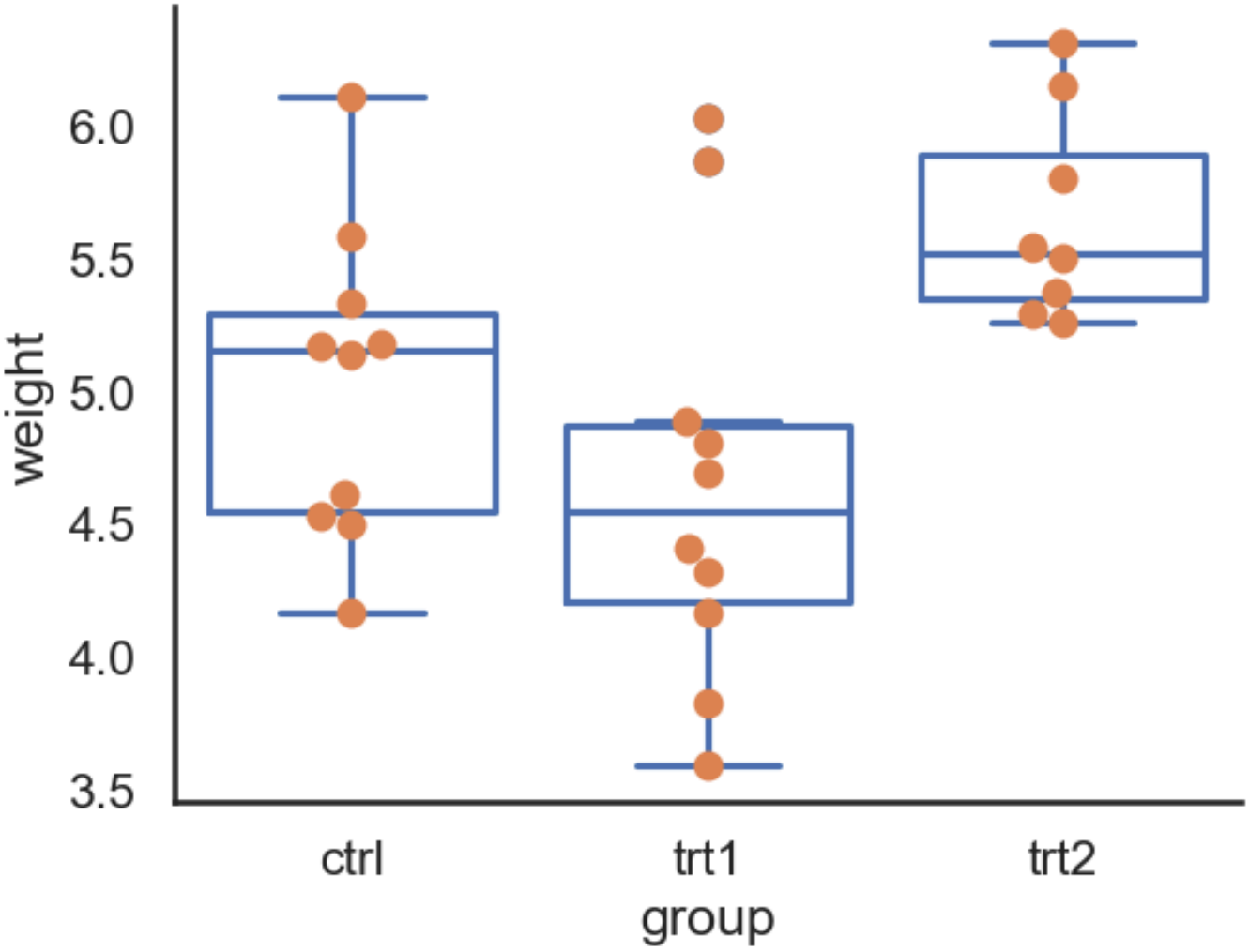
	weight	group
0	4.17	ctrl
1	5.58	ctrl
2	5.18	ctrl
3	6.11	ctrl
4	4.50	ctrl
5	4.61	ctrl
6	5.17	ctrl
7	4.53	ctrl
8	5.33	ctrl
9	5.14	ctrl
10	4.81	trt1

	weight	group
11	4.17	trt1
12	4.41	trt1
13	3.59	trt1
14	5.87	trt1
15	3.83	trt1
16	6.03	trt1
17	4.89	trt1
18	4.32	trt1
19	4.69	trt1
20	6.31	trt2
21	5.54	trt2
22	5.50	trt2
23	5.37	trt2
24	5.29	trt2

	weight	group
25	6.15	trt2
26	5.80	trt2
27	5.26	trt2

Plot the dry mass for each group

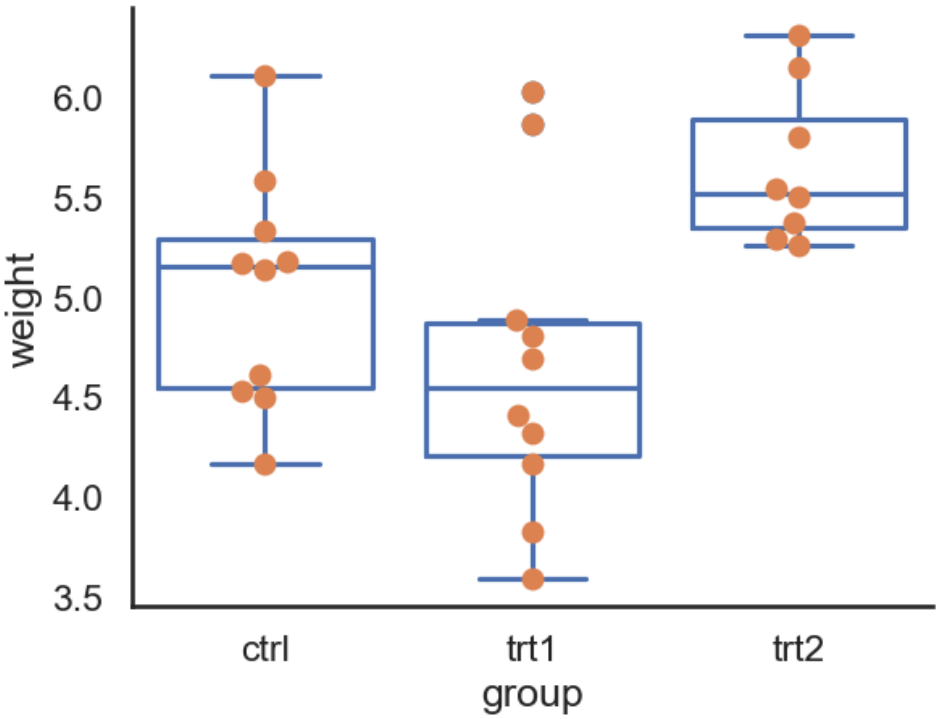
```
sns.boxplot(data=pg, x='group', y='weight', fill=None)
sns.swarmplot(data=pg, x='group', y='weight', size=10);
```



First do an ANOVA “omnibus” test

Check the compatibility of the data with the null hypothesis that all means are the same.

- If the data are compatible with the null ($P > 0.05$) the analysis stops there.
- If the data are incompatible with the null ($P < 0.05$) the analysis moves onto pairwise comparison with *t*-tests.



Two ways to code an ANOVA in python

anova_oneway

```
from statsmodels.stats.oneway import anova_oneway

test = anova_oneway(pg['weight'], pg['group'], use_var='equal')

print(test.df)
print(test.statistic)
print(test.pvalue)
```

```
(2.0, 25.0)
5.594501009822159
0.009818440749277929
```

Two ways to code an ANOVA in python

Linear model

```
formula = 'weight ~ group'
fit = ols(formula, pg).fit()
fit.summary().tables[0]
```

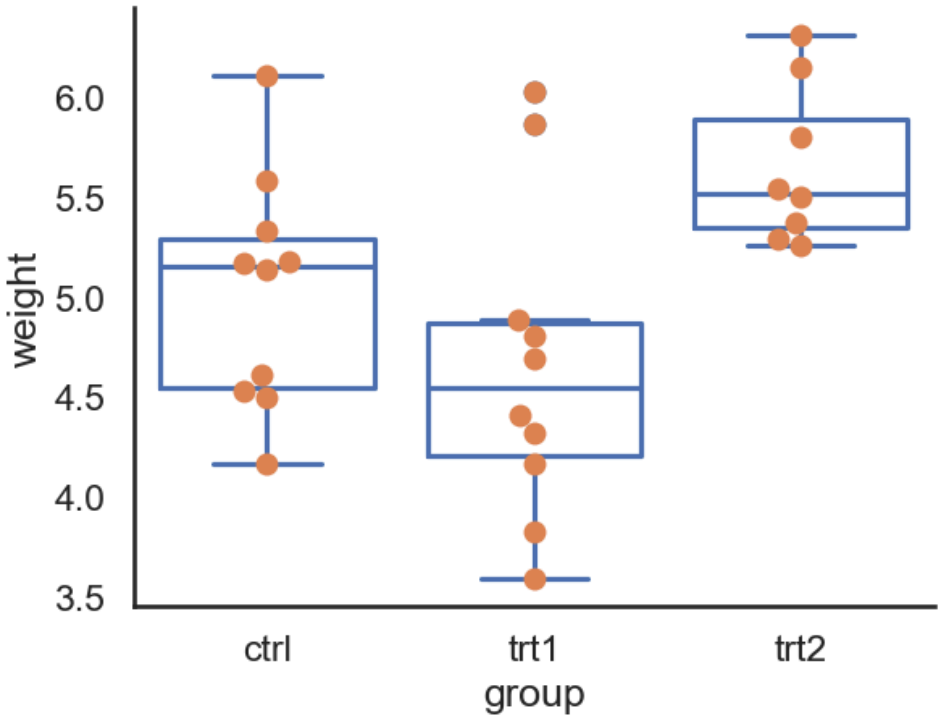
OLS Regression Results

Dep. Variable:	weight	**R-squared:**	0.309
Model:	OLS	**Adj. R-squared:**	0.254
Method:	Least Squares	**F-statistic:**	5.595
Date:	Thu, 29 Oct 2026	**Prob (F-statistic):**	0.00982
Time:	13:58:49	**Log-Likelihood:**	-25.078
No. Observations:	28	**AIC:**	56.16
Df Residuals:	25	**BIC:**	60.15
Df Model:	2		
Covariance Type:	nonrobust		

First do an ANOVA “omnibus” test

Both give the same result

- $F_{2,25} = 5.59$
- $P = 0.0098$
- Conclusion: At least one mean is different from at least one of the others.
- But which ones are different?



Second compare the treatments to the control

We need the raw *P*-values of the significance tests which we can obtain from the linear model

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	5.0320	0.198	25.374	0.000	4.624	5.440
group[T.trt1]	-0.3710	0.280	-1.323	0.198	-0.949	0.207
group[T.trt2]	0.6205	0.297	2.086	0.047	0.008	1.233

- Treatment 1 vs. control: $P_{\text{raw}} = 0.198$
- Treatment 2 vs. control: $P_{\text{raw}} = 0.047$

Adjust the raw *P*-values for multiple testing

```
from statsmodels.stats.multitest import multipletests

# extract the raw p-values from the linear model
p_raw = fit.pvalues.iloc[1:]

# apply adjustment using the Holm method
p_adj = multipletests(p_raw, method="holm")[1]
p_adj
```

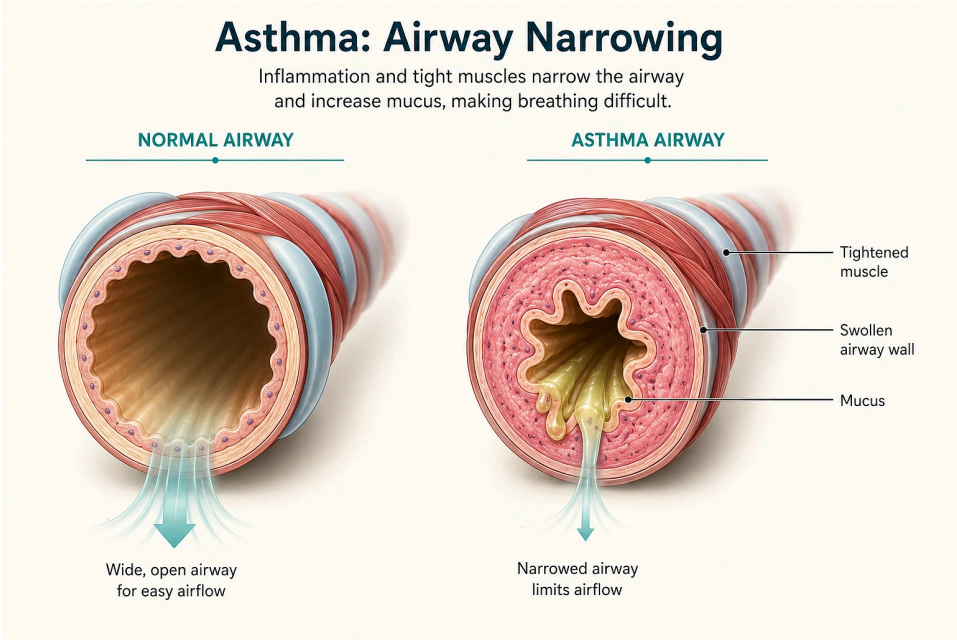
```
array([0.1978613, 0.0946938])
```

- Treatment 1 vs. control: $P_{adj} = 0.198$, not significant
- Treatment 2 vs. control: $P_{adj} = 0.095$, not significant

Conclusion: **neither fertiliser changed plant biomass**

By correcting for multiple testing we reduced the risk of making false claims.

Benjamini-Hochberg: differential gene expression



- Researchers wanted to better understand how the steroid, dexamethasone, suppresses inflammation in the lungs of people with asthma.
- They did this by identifying which genes of airway smooth muscle cells were differentially expressed when treated with dexamethasone.
- RNA-Seq, a high-throughput sequencing method, was used to quantify transcriptomic changes in thousands of genes.

Load and print the data

```
rna = pd.read_csv('../Datasets/airway_merged_data_c
rna.head(10)
```

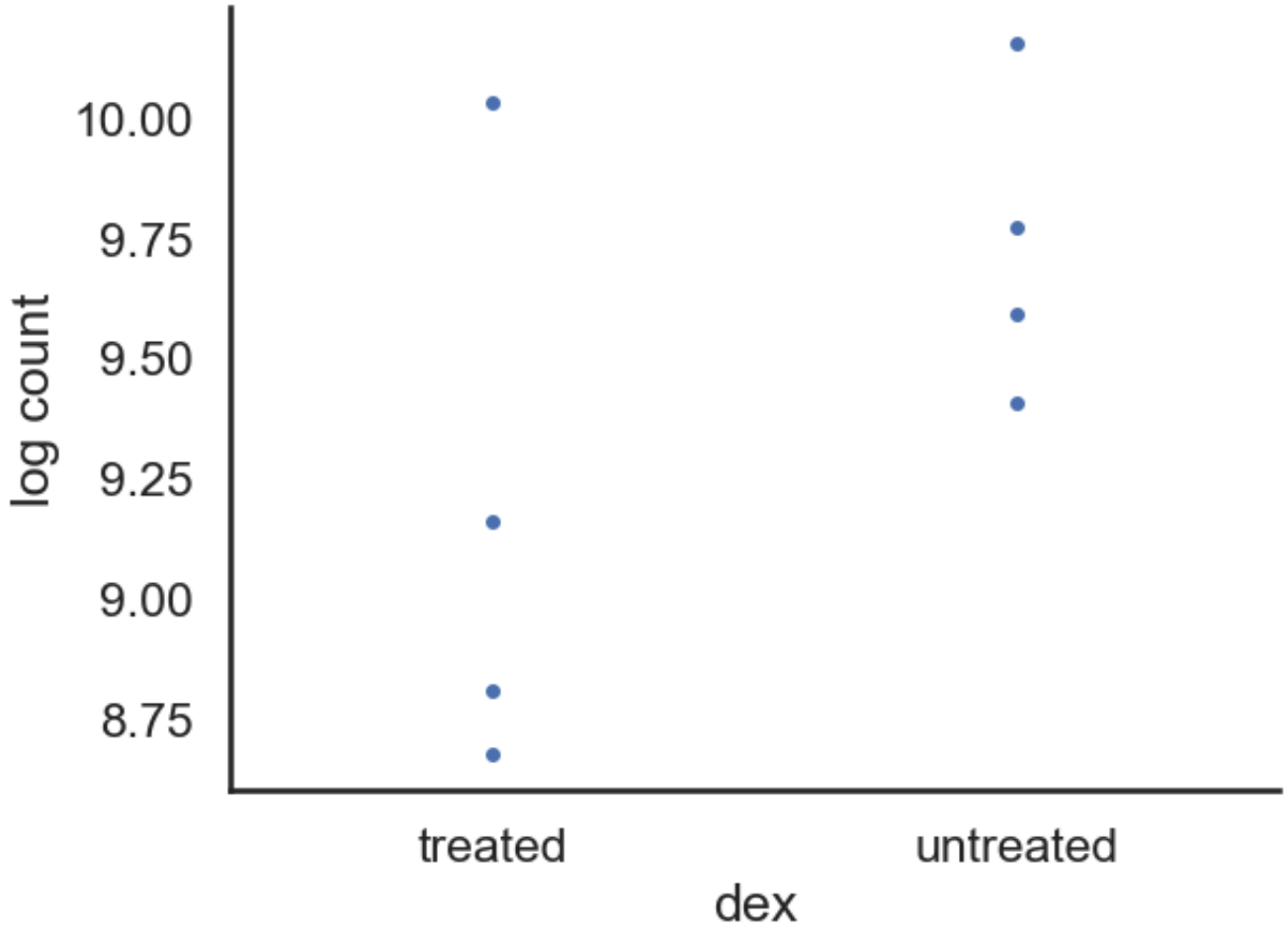
	gene_id	dex	cell_line	count
0	E000003	treated	S09	448
1	E000003	treated	S13	408
2	E000003	treated	S17	1047
3	E000003	treated	S21	572
4	E000003	untreated	S08	679
5	E000003	untreated	S12	873
6	E000003	untreated	S16	1138
7	E000003	untreated	S20	770
8	E000419	treated	S09	515
9	E000419	treated	S13	365

- There are 25,403 genes
- Four airway smooth muscle cell lines were treated with dexamethasone (S09, S13, S17, S21)
- Four airway smooth muscle cell lines were **not** treated with dexamethasone (S08, S12, S16, S20)
- The number of RNA short read sequences were counted in each cell line

Plot the read counts for a random gene

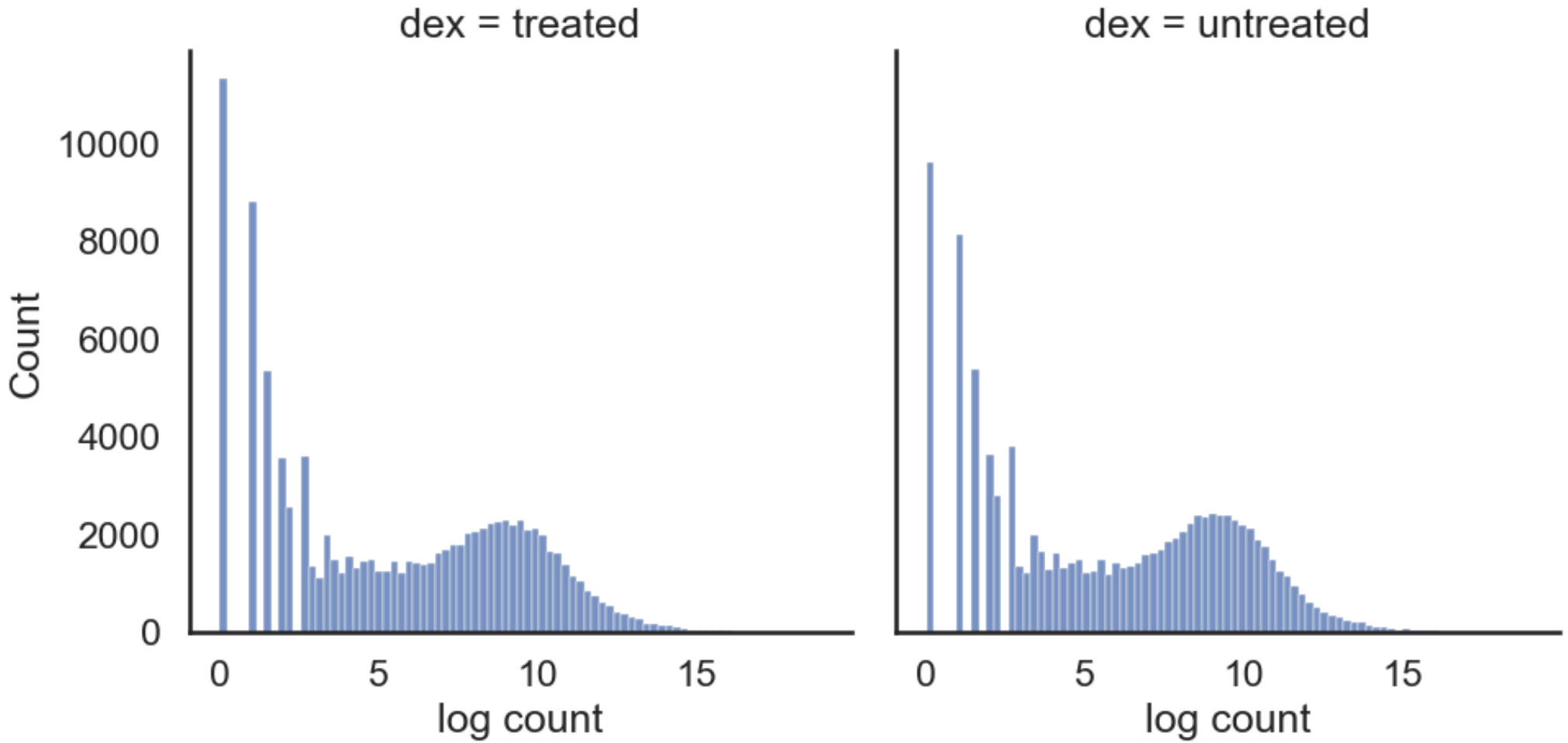
We have to log-transform RNA read counts (explained why in the workflow notebook)

```
sns.swarmplot(data=rna.query('gene_id == "E000003"'), y='log count', x='dex');
```



Plot the RNA read counts for all genes

```
sns.displot(data=rna, x='log count', col='dex');
```



First perform a *t*-test on each gene

- Null hypothesis: mean read counts are the same for treated and untreated cell lines for each gene
- There are 25,403 genes
- That's 25,403 *t*-tests
- If the null hypothesis were true for all of them we would get

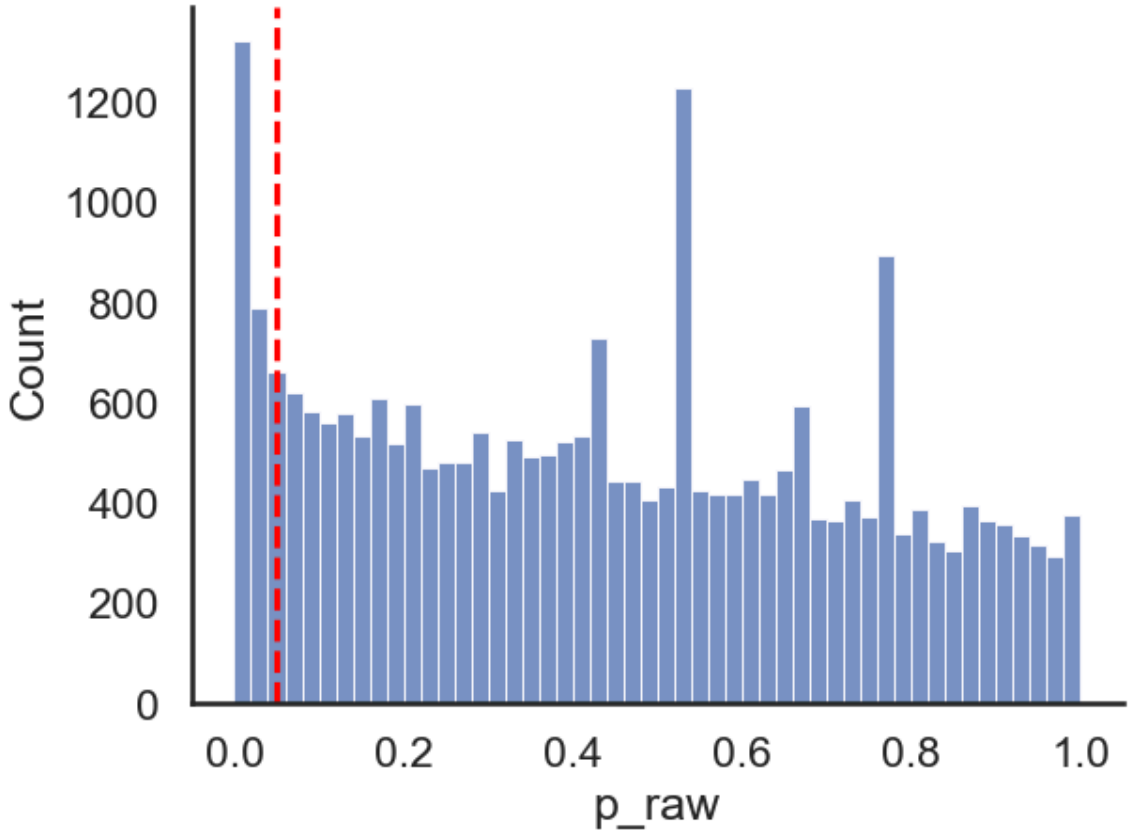
$$25{,}403 \times 0.05 = 1{,}207 \text{ false positives}$$

- We need to filter out as many false positives as possible

First perform a *t*-test on each gene

Distribution of raw *P*-values for 25,403 genes

The code to do this is in the workflow notebook.



There are 2,432 genes with $P < 0.05$

Adjust raw *P*-values using the Benjamini-Hochberg method

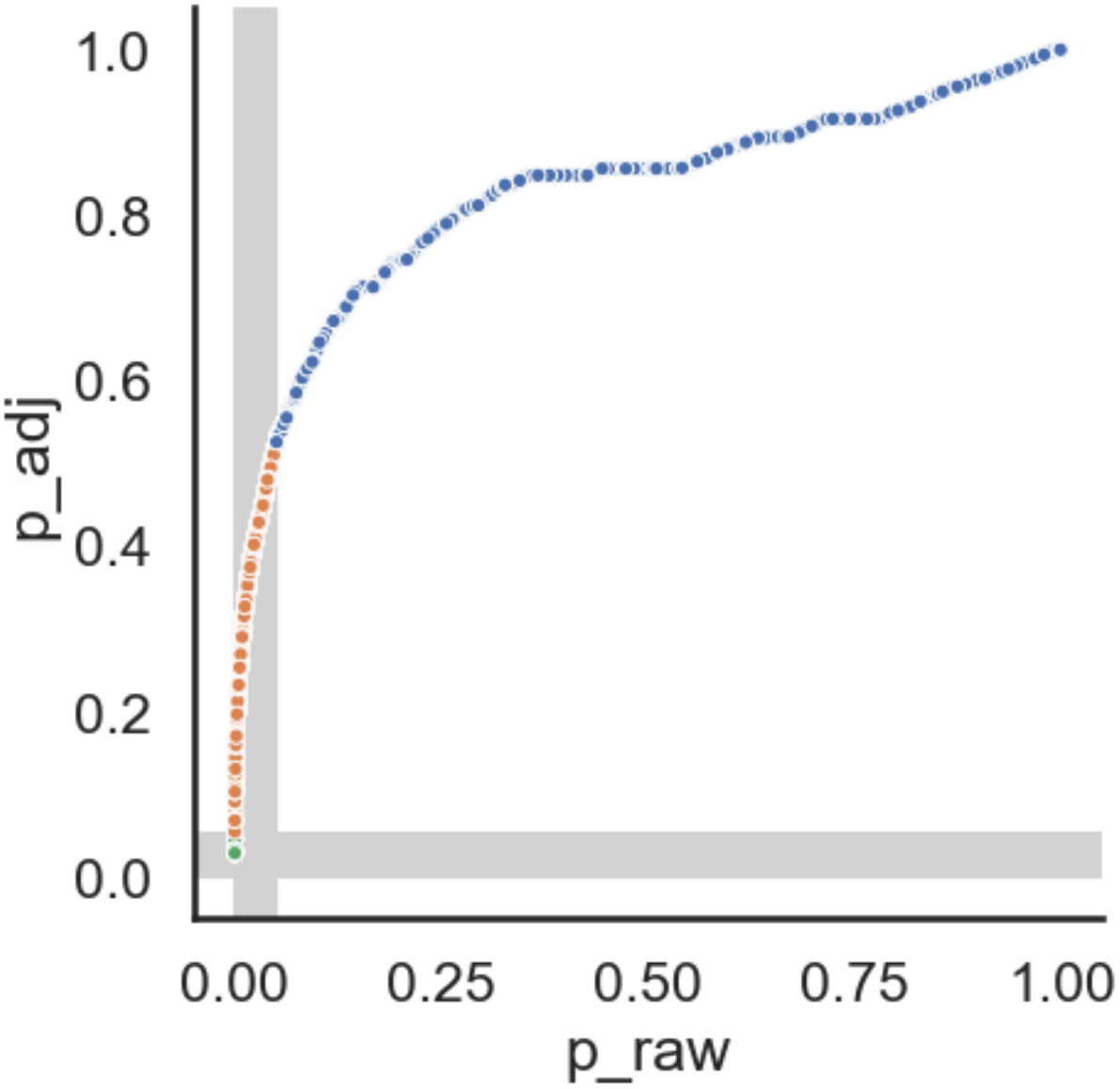
```
results_df['p_adj'] = multipletests(results_df['p_raw'], method='fdr_bh')[1]
results_df
```

	gene_id	p_raw	p_adj
0	E000003	0.154514	0.713029
1	E000419	0.925824	0.972746
2	E000457	0.527302	0.856147
3	E000460	0.443651	0.856147
4	E000971	0.598771	0.881235
...	...	...	...
25398	E273477	0.698030	0.907591
25399	E273483	0.473038	0.856147
25400	E273486	0.510124	0.856147

	gene_id	p_raw	p_adj
25401	E273487	0.029086	0.428529
25402	E273488	0.874195	0.955385

25403 rows × 3 columns

What Benjamini-Hochberg does to *P*-values



Print out the genes with significant adjusted *P* values

	gene_id	p_raw	p_adj
291	E013297	0.000009	0.028547
13575	E187193	0.000015	0.028547
11471	E173110	0.000014	0.028547
8664	E155962	0.000006	0.028547
8347	E152583	0.000003	0.028547
7378	E143494	0.000007	0.028547
6027	E134253	0.000005	0.028547
5284	E128285	0.000012	0.028547
4729	E122877	0.000011	0.028547
24138	E267683	0.000012	0.028547
2263	E101347	0.000004	0.028547
2798	E105516	0.000013	0.028547
2260	E101342	0.000005	0.028547
10153	E165995	0.000021	0.032413
4476	E120129	0.000026	0.032413

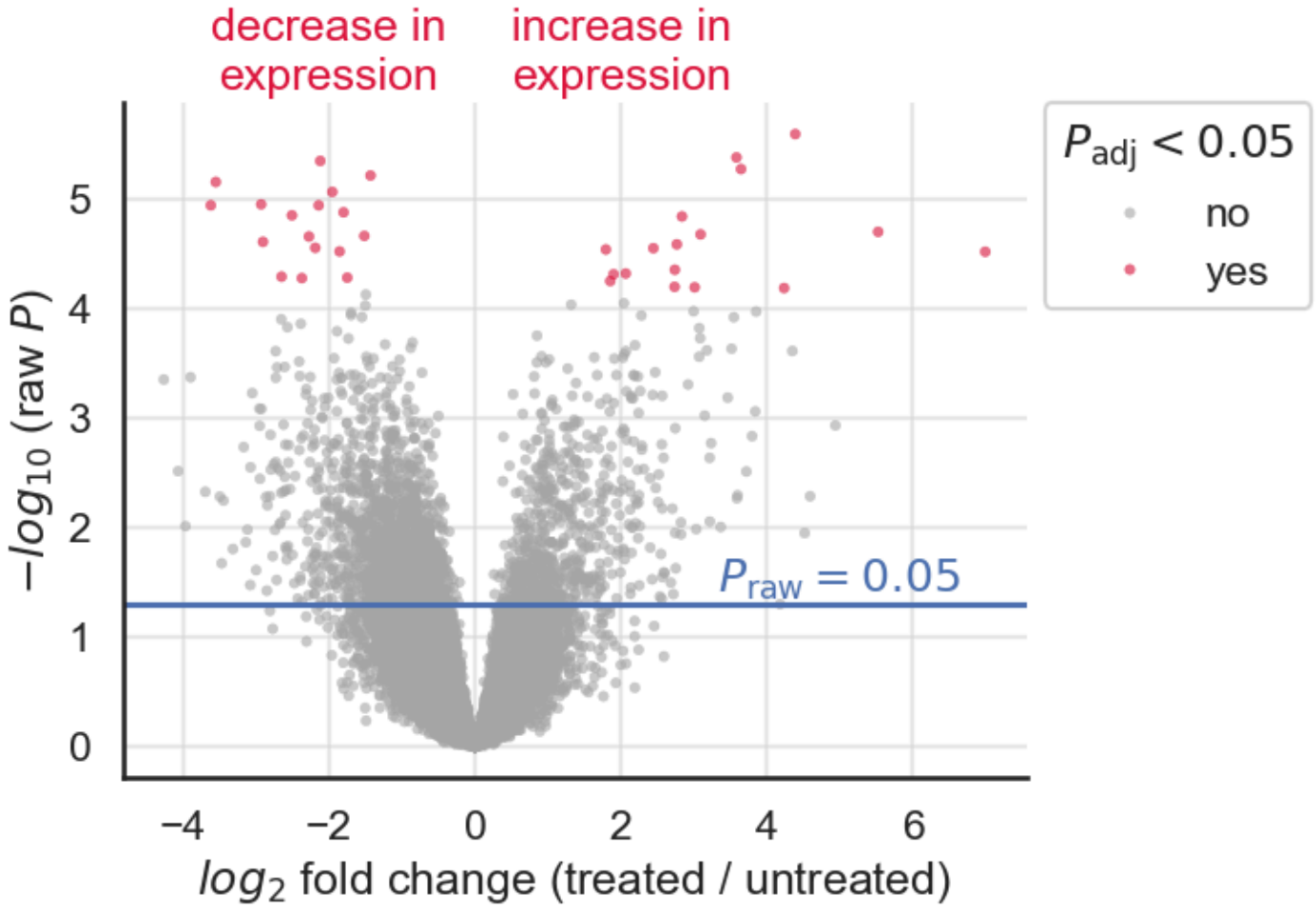
There are 34 such genes

- The Benjamini-Hochberg method says that, of all the significant genes discovered, on average 5% of them are expected to be false positives.
- There are 34 discovered genes. 5% of that is 1.7 genes. So we might expect of all the genes discovered, 1 or 2 of them are false positives.
- That is much better than using the raw *P*-values which gave us 2,432 genes with significantly changed expression.

16151	E215386	0.000022	0.032413
8792	E157214	0.000029	0.032413
7677	E145777	0.000025	0.032413
21292	E250978	0.000020	0.032413
2496	E103196	0.000028	0.032413
20531	E243742	0.000028	0.032413
5641	E131242	0.000022	0.032413
3328	E109906	0.000031	0.032413
2408	E102524	0.000030	0.032413
14589	E198624	0.000045	0.045136
597	E049246	0.000053	0.045136
13283	E185432	0.000048	0.045136
4351	E119138	0.000049	0.045136
11456	E172986	0.000053	0.045136
9583	E163491	0.000052	0.045136
9393	E162614	0.000057	0.046423

Volcano plot of log-fold change in gene expression against significance

This is explained in more detail in the workflow notebook.



Class 12 notebook

Three questions on mutliple testing:

1. Compare multiple treatments to a control (Holm)
2. Compare each treatment to every other treatment (Holm)
3. Multiple individual tests (Benjamini-Hochberg)

Workflow notebooks

1. **workflow_plant_growth.ipynb**: Multiple tests against a control
2. **workflow_iris_sepal_width.ipynb**: Pairwise tests between all treatments
3. **workflow_gene_expression.ipynb**: Multiple individual tests